



GE Medical Systems

Technical Publication

Direction 2319916-100

Revision 0

**LightSpeed Ultra
8 Slice Upgrade Procedure**

OPERATORS DOCUMENTATION

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REVISION HISTORY

REVISION HISTORY

Revision	Date	Reason for change
A	11/9/2001	First Draft
B	11/13/2001	Corrections from first review
0	11/14/01	Initial Release

TABLE OF CONTENTS

Legal Notes	3
TRADEMARKS	3
OMISSIONS & ERRORS	3
COPYRIGHTS	3
Important Precautions	4
LANGUAGE	4
LANGUAGE	5
Revision History	7
Table of Contents	8
Section 1 - Introduction	9
Section 2 - System Requirements	10
SECTION 2.1 HARDWARE REQUIRED TO SUPPORT 8 SLICE SCANNING	10
SECTION 2.2 SOFTWARE REQUIRED TO SUPPORT 8 SLICE SCANNING	10
Section 3 - 8 Slice Option Installation	11
SECTION 3.1 - INSTALLATION STEERING GUIDE	11
SECTION 3.2 - INSTALLATION PROCEDURE	12
<i>Section 3.2.1 - Verify System Compatibility</i>	<i>12</i>
<i>Section 3.2.2 - Archive Patient Images</i>	<i>12</i>
<i>Section 3.2.3 - Software Load From Cold</i>	<i>12</i>
<i>Section 3.2.4 - Update Service Protocols</i>	<i>12</i>
<i>Section 3.2.5 - Preliminary Calibration</i>	<i>13</i>
<i>Section 3.2.6 - Verify 8 slice Operation</i>	<i>13</i>
<i>Section 3.2.7 - Detailed Calibration of 8 slice Modes</i>	<i>13</i>
<i>Section 3.2.8 - System Checkout and Save State</i>	<i>14</i>
Appendix A - Image Series (System Image Performance Verification)	15
48CM PHANTOM IMAGE SERIES IMAGE PERFORMANCE VERIFICATION	15
<i>Acquiring the 48cm Phantom Image Series</i>	<i>15</i>
<i>Brightness Uniformity and Noise Image Performance Verification</i>	<i>17</i>
<i>48cm Phantom Image Series Image Performance Failure Recommended Actions</i>	<i>20</i>
20CM QA PHANTOM IMAGE SERIES IMAGE PERFORMANCE VERIFICATION	21
<i>Acquiring the 20cm QA Phantom Image Series</i>	<i>21</i>
<i>Brightness Uniformity and CT# Performance Verification</i>	<i>23</i>
<i>20cm QA Phantom Image Series Image Performance Failure Recommended Actions</i>	<i>26</i>

SECTION 1 - INTRODUCTION

The 8 Slice Upgrade kit is intended to be used on the installed base of LightSpeed Plus (MDAS) scanners. This upgrade kit **will not work with LightSpeed Plus (SDAS)** scanners. Through software and hardware changes the (MDAS) system will become enabled to scan and reconstruct 8 slice images in the following two modes:

8 X 1.25 mm

8 X 2.5 mm

All of the 4 Slice functionality will remain the same as the LightSpeed Plus.

Detail Calibrations for 8 slice will add approximately 1 hour of time to the detail calibration cycle time.

Table 1 details the materials shipped as a part of the upgrade kit B7858SM (2308199).

Part Number	Part Description
See next 5 items	Option MOD including the options listed below
2316996	8 Slice Option
2284205	Smart Speed Option
2305988	Thin Twin Helical Option
2305990	Helical Tilt Option
2305991	Recon Plus Option
2305095-3	LightSpeed Ultra Serialized Software Collector
2305091-3	Software Collector
2305087	IRIX64 VER 6.5.11M P2 DISK 1
2305088	IRIX64 VER 6.5.11 Pa DISK 2
2305089	IRIX64 VER 6.5.11 Pb DISK 3
2305090-3	LightSpeed Ultra Application Software Ver 301
2296447-100	LightSpeed Ultra Applications Tips & Workarounds
2304119-248	LightSpeed Ultra Operator Documentation (CDROM)
2302180-100	LightSpeed Ultra Technical Reference Manual
2296436-100	LightSpeed Ultra Software Load Procedure
2319916-100	LightSpeed Ultra 8 Slice Upgrade Procedure

Table 1 Required Material 8 Slice Upgrade Kit (2308199)

SECTION 2 - SYSTEM REQUIRMENTS

This upgrade is only intended to operate on LightSpeed Plus systems installed with an MDAS detector system. This system can be identified by the gantry model number (2281177).

SECTION 2.1

HARDWARE REQUIRED TO SUPPORT 8 SLICE SCANNING

LightSpeed Ultra (H3) scanners built in 2001 all contain 48 DAS Converter Boards.

- MDAS Assembly H3 8 Slice Part Number 2272768 No hardware required

SECTION 2.2

SOFTWARE REQUIRED TO SUPPORT 8 SLICE SCANNING

2305095-3	LightSpeed Plus/Ultra Serialized Software
2316996	8 Slice Option Key

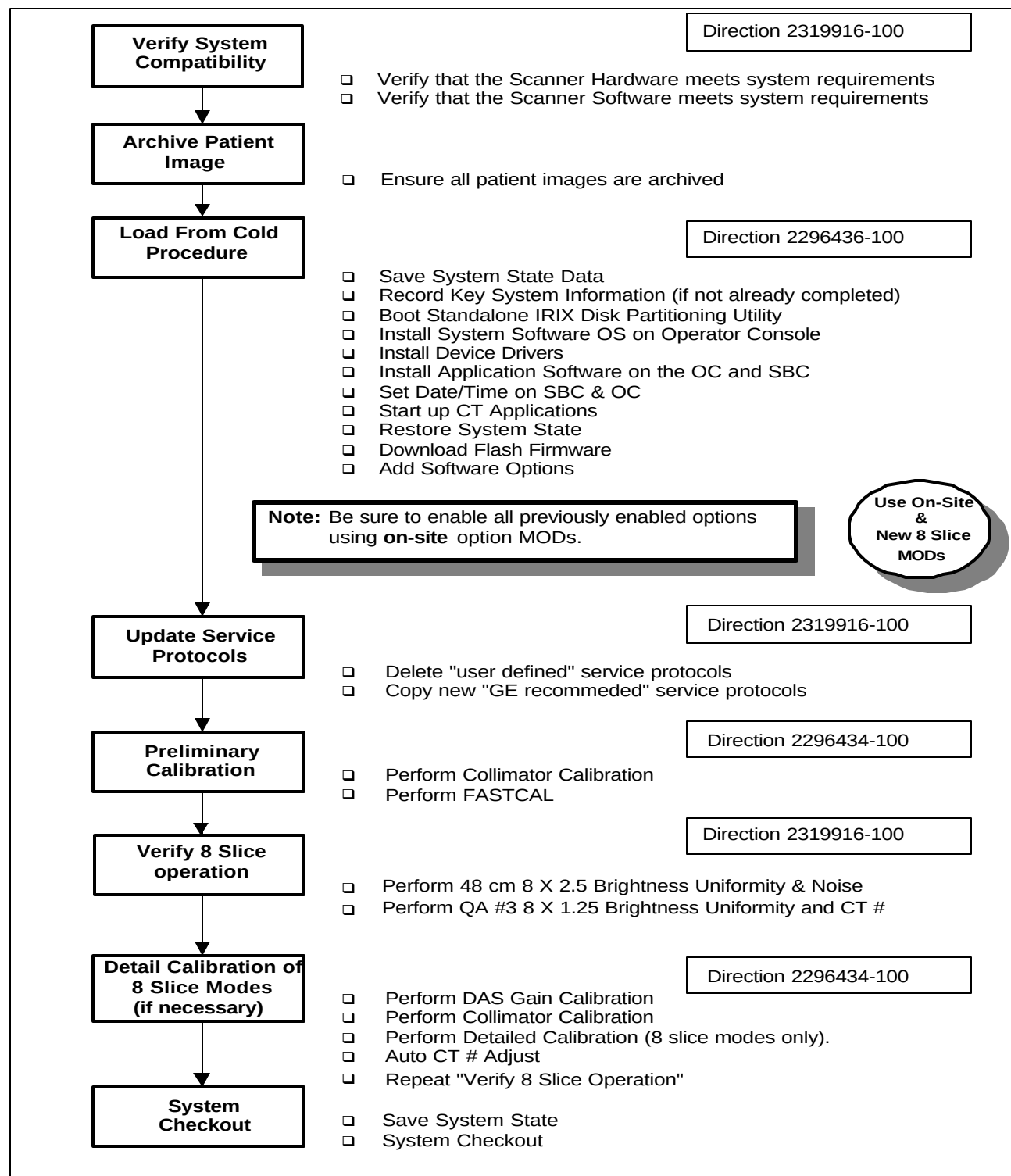
Opening the Service Desktop and selecting the HOME tab will display system information. In the section labeled "Current System Status" you will find the "Application Software".

The version displayed for the LightSpeed Plus/Ultra **8 Slice** M3 is:

301.2_2.9C_0628_H3M3

SECTION 3 - 8 SLICE OPTION INSTALLATION

SECTION 3.1 - INSTALLATION STEERING GUIDE



SECTION 3.2 - INSTALLATION PROCEDURE

SECTION 3.2.1 - VERIFY SYSTEM COMPATIBILITY

1. Compare your system hardware to the system hardware requirements listed in Section 2.
2. If your gantry model number is **not** 2281177, **STOP** and contact sales representative.
3. If your gantry model number is 2281177, continue to the next step.
4. Compare your system software to the system software requirements listed in Section 2.

Open the Service Desktop and select the HOME tab to display system information. In the section labeled "**Current System Status**" you will find the "**Application Software**".

- The version displayed for the LightSpeed Plus/Ultra **4 Slice** M3 (does not support 8 Slice) is:
300.2.2.9C_0628_H34M3
- The version displayed for the LightSpeed Plus/Ultra **8 Slice** M3 is:
301.2.2.9C_20010628_H3M3

5. If the system does not have version 301 or higher software, verify that you have software collector 2305095-3 and Options MOD listed in Table 1, available for software load.

SECTION 3.2.2 - ARCHIVE PATIENT IMAGES

Verify that the customer has archived all patient data.

SECTION 3.2.3 - SOFTWARE LOAD FROM COLD

Perform the Load From Cold procedure as detailed in Chapter 1 of "LightSpeed Ultra Software Installation Procedures" (2296436-100).

NOTICE: A Flash Download must be performed after the 8 Slice option is loaded.

SECTION 3.2.4 - UPDATE SERVICE PROTOCOLS

This section details the process for updating protocols that were written to the system as part of the Restore STATE process. A new set of Service Protocols have been created for the support of 8 Slice testing. These have been added to the system as a part of the LFC process. However, they are not displayed because the Restore STATE process writes the previous customer protocols over them. A new feature allows us to selectively copy "GE recommended" protocols into the displayed "User Defined" protocols.

1. With System Applications running on the scanner, select the **Copy Protocols** button from the top right corner of the right monitor. This will open a GUI that is used for updating the protocols.
2. Cursor select the patient image (adult). This will change the patient image to a baby and display the protocols associated with this image.
3. Expand the list under the selection "Infant Upper Extremity" on the User Defined side of the GUI by clicking on the words for this selection. This will display all of the service protocols associated with "Infant Upper Extremity".

4. Delete each of the protocols listed under the User Defined/Infant Upper Extremity directory by selecting each protocol and pressing the **Delete** button.

NOTICE: To copy a protocol from the GE Recommended to the User Defined side of the GUI you must first select the User Defined directory that you wish to copy the protocol into. Then select the GE Recommended protocol to be copied.

5. Copy each of the protocols found in the GE Recommended/Infant Upper Extremity directory to the User Defined/Infant Upper Extremity directory by selecting each protocol and pressing the **Copy** button.
6. Expand the list under the selection User Defined "Infant Miscellaneous" by clicking on the words. This will display all of the service protocols associated with "Infant Miscellaneous".
7. Delete each of the protocols listed under the User Defined/Infant Miscellaneous directory by selecting each protocol and pressing the **Delete** button..

NOTICE: To copy a protocol from the GE Recommended to the User Defined side of the GUI you must first select the User Defined directory that you wish to copy the protocol into. Then select the GE Recommended protocol to be copied.

8. Copy each of the protocols found in the GE Recommended/Infant Miscellaneous directory to the User Defined/Infant Miscellaneous directory by selecting each protocol and pressing the **Copy** button.

SECTION 3.2.5 - PRELIMINARY CALIBRATION

Every LightSpeed Ultra 4 slice scanner was calibrated in manufacturing as if it was an 8 slice scanner. With this assumption, this section will perform the minimal calibrations required to assure 8 slice functionality.

1. Perform Collimator Calibration by selecting Scanner Utilities on bottom of the left monitor then selecting Collimator Calibration from the displayed selections.
2. Perform FASTCAL by selecting Daily Prep on the bottom of the left monitor, then selecting FASTCAL from the displayed selections.

NOTICE: You may get a pop-up indicating: "System detects an alignment or change was performed on the System". If this happens, press the Ok button and continue with the FASTCAL.

SECTION 3.2.6 - VERIFY 8 SLICE OPERATION

The functionality of your 8 slice option can be tested by performing the detailed testing outlined in Appendix A of this document. This process uses the 48cm Body Phantom and the 20cm QA phantom. Two acquisitions are performed and then the images are processed and evaluated per the instructions in Appendix A.

If the evaluation process meets the specifications detailed in Appendix A, skip Section 3.2.7 and go directly to Section 3.2.8.

If the specifications are not met in the testing performed in Appendix A, proceed to Section 3.2.7 and perform a Detailed Calibration.

SECTION 3.2.7 - DETAILED CALIBRATION OF 8 SLICE MODES

1. Perform DAS Gain Calibration by selecting Scanner Utilities on bottom of the left monitor then selecting DAS Gain Calibration from the displayed selections.

2. Perform Collimator Calibration by selecting Scanner Utilities on bottom of the left monitor then selecting Collimator Calibration from the displayed selections.
1. Perform Detailed Calibration by selecting Scanner Utilities on bottom of the left monitor then selecting Detailed Calibration from the displayed selections. Highlight the 8 Slice modes, customer preferred KVs, all SFOVs, small and large focal spot.
2. Perform Auto CT Number Adjust by selecting Scanner Utilities on bottom of the left monitor then selecting Auto CT Number Adjust from the displayed selections.
3. Verify 8 Slice operation by performing the tests detailed in Appendix A of this document.

SECTION 3.2.8 - SYSTEM CHECKOUT AND SAVE STATE

1. Refer to the LFC Manual, and perform a Save System State. This will save the new calibrations as well as the customer protocols.
2. Refer to the LFC Manual, and perform the system performance checks as described in the manual. Ensure all networking and filming processes are still functional.
3. Install Option rating plate to the rear of Console near existing rating plate.
4. Complete and return Red Installation Product Locator Cards.

APPENDIX A - IMAGE SERIES (SYSTEM IMAGE PERFORMANCE VERIFICATION)

48CM PHANTOM IMAGE SERIES IMAGE PERFORMANCE VERIFICATION

ACQUIRING THE 48CM PHANTOM IMAGE SERIES

1. Mount the Phantom Holder on the head-end of the table.
2. Mount the 48cm Phantom on the Phantom Holder.
3. Align, level, & center the 48cm Phantom.
 - Align phantom using the internal laser lights.
 - Level phantom using bubble level and the Z Axis knob on the Phantom Holder.
 - Center phantom using the **Center Phantom** procedure in the left head **Scanner Utilities** selection and the X and Y Axis knobs on the Phantom Holder.
4. Set up the system to scan a single, 4 image, 48cm Phantom series.

MANUAL SCAN PROTOCOL SETUP

Refer to Table 1 and set-up an Axial scan with the parameters shown.

Table 1 LightSpeed QX/I 48cm Phantom Image Series Scan Parameters

Series Description	Scan Type	Start Loc.	End Loc.	Total # of Images	Thick Speed	Interval (mm)	Gantry Tilt	SFOV	kV	mA	Total Exposure Time	DFOV (cm)	Recon Type
48 Lg Series 1	Axial Full 1.0 sec.	17.50	S7.50	8	2.5 8i	0.00	S0.0	Large	120	340	1.0 sec.	50.0	Std

AUTO SCAN PROTOCOL SETUP

- a. On the Exam Rx desktop, select **New Patient**.
 - b. Type the following entries in the two listed **Patient Information** fields:
 - **Patient ID:** Service
 - **Name:** 48cm Phantom
 - c. From the Protocol display, click on the infant box.
 - d. On the Infant display window, click on the area below the infant's left foot to display the **miscellaneous** menu.
 - e. Click on the **20:10 Image Series 48 cm** button.
 - f. Use the first series.
5. Set internal Landmark.

6. Acquire a single, 8 image, scan of the 48 cm Phantom.

BRIGHTNESS UNIFORMITY AND NOISE IMAGE PERFORMANCE VERIFICATION

1. Select the 48cm exam acquired in the previous section.
 - a. From the Global Control Palette, click on the **Image Works** Desktop.
 - b. From the Image Works Desktop, select the **Image Works Browser** window.
 - c. Select the exam and series acquired in the previous section.
 - d. Select the **Viewer** button on the Image Works Browser window. Set up the viewer window for four-image viewing.
2. Build a 45 x 45 pixel reference ROI Box using the Image Works Viewer tools.
 - a. Click on the grid button to place a grid on the first image.
 - b. Click on the **Measure** button and select the box ROI icon.
 - c. Adjust the size of the ROI box to be a square 45 mm x 45 mm (2025 mm²) box. Tolerance: 45 mm +/- 5 mm (1600 mm² to 2500 mm²).

NOTICE: If required, magnify the image to adjust to proper dimensions. See Figure 1 for additional ROI size and placement information.

3. Collect Mean and Standard deviation values for five reference ROI Box positions on the eight 48cm Phantom images.

NOTICE: If required, magnify the image to adjust to proper dimensions. See Figure 1 for additional ROI placement information.

- a. Position the reference ROI Box built in step 2 directly over the center of the image using the grid crosshairs as a guide.
- b. Type "**prop a**" in the Image Works Accelerator Line followed by <Return> to propagate Box # 1 ROI (size and position) on the remaining images in the series.
- c. Click on the **Measure** button and select the box ROI icon to display Box # 2. The system places an ROI box labeled # 2 at the center of the image with the exact same dimensions as Box # 1.
- d. Re-position Box # 2 to the left center portion on the first image (Box # 2 position in Figure 1).
- e. Type "**prop a**" in the Image Works Accelerator Line followed by <Return> to propagate Box # 2 ROI (size and position) on the remaining images.
- f. Repeat steps 3.c. through 3.e. for each of the remaining ROI box positions shown in Figure 1 (Box # 3 through 5).
- g. Record the Mean and Standard Deviation values of the eight images in the series for each of the five box positions in Table 2.

NOTICE: Each image can only display text for the mean, standard deviation, and box area for three images at a time. To view the data for a particular box, select the box on the image and the system displays the data for the box number selected.

Figure 1 **48cm Phantom Brightness Uniformity & Noise Measurement – Building**
Placing Reference ROI Boxes

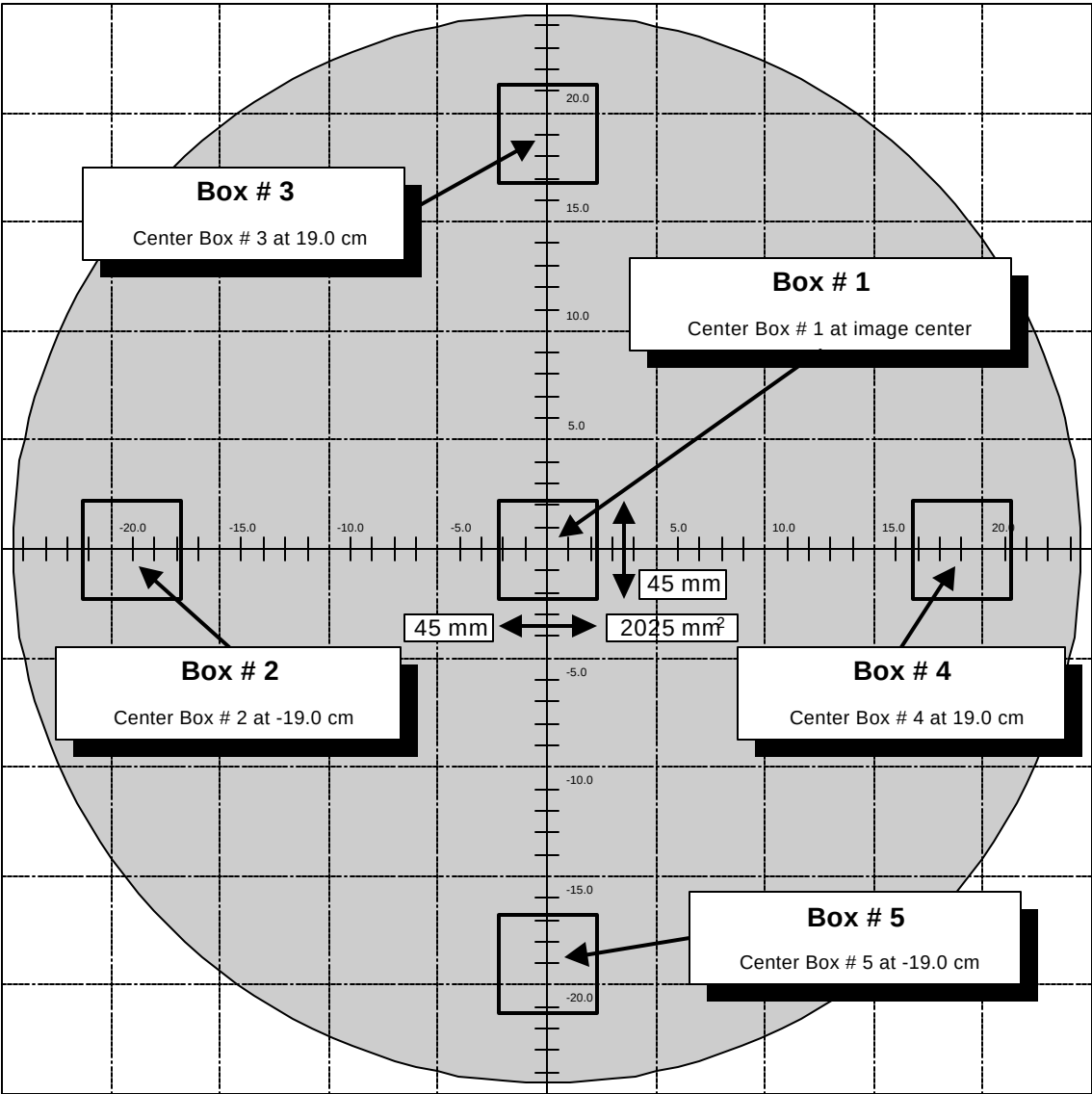


Table 2 **48cm Phantom CT# Brightness Uniformity & Noise Image Performance Worksheet**

Image	Box 1 (Center)	Box 2 (Left Center)		Box 3 (Top Center)		Box 4 (Right Center)		Box 5 (Bottom Center)		Outside Boxes Avg	
	Means (Xc)	Means	Std dev	Means	Std dev	Means	Std dev	Means	Std dev	Means (AvXo)	Std dev (AvSDo)
1											
2											
3											
4											
5											
6											
7											
8											

Box Size = 1600 mm² to 2500 mm²
 45 mm (+/- 5 mm) x 45 mm (+/- 5 mm)
 45 (+/- 4 pixels) x 45 (+/- 4 pixels)

Box Positions: Box 1 = 0 mm x 0 mm
 Box 2 = 0 mm x -190 mm
 Box 3 = 190 mm x 0 mm
 Box 4 = 0 mm x 190 mm
 Box 5 = -190 mm x 0 mm

Table 3 **48cm Phantom CT# Brightness Uniformity & Noise Row Performance Worksheet**

Row	Image s	Center Box Means (Xc)	Outer Boxes Means Averages (AvXo)	Outer Boxes Means Averages minus Center Box Means (AvXo – Xc)	Outer Boxes Standard Deviation Averages (AvSDo)	Comments
4A	1					
3A	2					
2A	3					
1A	4					
1B	5					
2B	6					
3B	7					
4B	8					
Specifications		N.A.	N.A.	+/- 8.5	< 70	

4. Calculate the Brightness Uniformity and Noise values for each image in the series, record values in Table 3, and compare to specifications.
 - a. Calculate and record the average means and average standard deviation values for the four outside Boxes (Boxes 2 through 5) for each of the images and record in Table 2.
 - b. Record the center box (Box 1) means values (X_c) for each image in Table 3.
 - c. Record the outside boxes (Box 2 through 5) means average values (AvX_o) and standard deviation average values ($AvSD_o$) for each image in Table 3.
 - d. Calculate the Brightness Uniformity ($AvX_o - X_c$) value for each image and record in Table 3.
 - e. Verify Brightness Uniformity ($AvX_o - X_c$) value and Noise value ($AvSD_o$) for each image meets specifications listed in Table 2.
 - f. Record the 48cm Phantom Brightness Uniformity ($AvX_o - X_c$) value and Noise value ($AvSD_o$) for each image in the HHS Record Tables.

48CM PHANTOM IMAGE SERIES IMAGE PERFORMANCE FAILURE RECOMMENDED ACTIONS

SPECIFICATIONS:

Each Row (4A, 3A, 2A, 1A, 1B, 2B, 3B, and 4B) of the series must pass 48cm Brightness Uniformity and Noise (for the first series scan parameters) specifications:

- **$AvX_o - AvX_c$:** < +/- 8.5
- **$AvSD_o$:** < 70.0

RECOMMENDED RECOVERY:

If the specifications are not met, perform DAS Gain Calibration, Collimator Calibration, Detailed Calibration and Auto CT Number Adjust as detailed in Section 3.2.7 of this document. Repeat this procedure to verify 48cm Phantom Image Performance.

20CM QA PHANTOM IMAGE SERIES IMAGE PERFORMANCE VERIFICATION

ACQUIRING THE 20CM QA PHANTOM IMAGE SERIES

1. Mount the Phantom Holder on the head-end of the table.
2. Mount the 20cm QA Phantom on the Phantom Holder.
3. Align, level, & center the 20cm QA Phantom.
 - Align black line on phantom using the internal laser lights.
 - Level phantom using bubble level and the Z Axis knob on the Phantom Holder.
 - Center phantom using the **Center Phantom** procedure in the left head **Scanner Utilities** selection and the X and Y Axis knobs on the Phantom Holder.
4. Set up the system to scan the QA#3 position on the 20cm QA Phantom image series.

MANUAL SCAN PROTOCOL SETUP

Refer to Table 4 and set-up an Axial scan with the parameters shown.

Table 4 LightSpeed QX/I 20cm QA Phantom Image Series Scan Parameters

Series Description	Scan Type	Start Loc.	End Loc.	Total # of Images	Thick Speed	Interval (mm)	Gantry Tilt	SFOV	kV	mA	Total Exposure Time	DFOV (cm)	Recon Type
QA#3	Axial Full 2.0 sec.	S58.10	S66.85	8	1.25 8i	0.00	S0.0	Small	120	200	2.0 sec.	25.0	Std

AUTO SCAN PROTOCOL SETUP

- a. On the Exam Rx desktop, select **New Patient**.
 - b. Type the following entries in the listed **Patient Information** following fields:
 - **Patient ID:** Service
 - **Name:** 20cm QA#3 Phantom Image Series
 - c. From the Protocol display, click on the infant box.
 - d. On the Infant display window, click on the area below the infant's left foot to display the **miscellaneous** menu.
 - e. Click on the **20:12 Image Series QA** button.
 - f. When the scan prescription appears on the left monitor, skip to the QA#3 / small / 8X1.25 series (displayed in the series description box) by pressing the next series button.
5. Set internal Landmark.

6. Acquire the QA#3 position of the 20cm QA Phantom image series.

BRIGHTNESS UNIFORMITY AND CT# PERFORMANCE VERIFICATION

1. Select the 20cm QA Phantom exam acquired in the previous section.
 - a. From the Global Control Palette, click on the **Image Works** Desktop.
 - b. From the Image Works Desktop, select the **Image Works Browser** window.
 - c. Select the exam and series acquired in the previous section.
 - d. Select the **Viewer** button on the Image Works Browser window. Set up the viewer window for four-image viewing.
2. Build a 31 x 31 pixel reference ROI Box using the Image Works Viewer tools.
 - a. Click on the grid button to place a grid on the first image.
 - b. Click on the **Measure** button and select the box ROI icon.
 - c. Adjust the size of the ROI box to be a square 15 mm x 15 mm (225 mm²) box. Tolerance: 15 mm +/- 1 mm (196 mm² to 256 mm²).

NOTICE: If required, magnify the image to adjust to proper dimensions. See Figure 2 for additional ROI size and placement information.

3. Collect Mean values for five reference ROI Box positions on the eight 20cm QA Phantom images.

NOTICE: If required, magnify the image to adjust to proper dimensions. See Figure 2 for additional ROI placement information.

- a. Position the reference ROI Box built in step 2 directly over the center of the image using the grid crosshairs as a guide.
- b. Type "**prop a**" in the Image Works Accelerator Line followed by <Return> to propagate Box # 1 ROI (size and position) on the remaining images in the series.
- c. Click on the **Measure** button and select the box ROI icon to display Box # 2. The system places an ROI box labeled # 2 at the center of the image with the exact same dimensions as Box # 1.
- d. Re-position Box # 2 to the left center portion on the first image (Box # 2 position in Figure 1).
- e. Type "**prop a**" in the Image Works Accelerator Line followed by <Return> to propagate Box # 2 ROI (size and position) on the remaining images.
- f. Repeat steps 3.c. through 3.e. for each of the remaining ROI box positions shown in Figure 2 (Box # 3 through 5).
- g. Record the Mean values of the eight images in the series for each of the five box positions in Table 5.

NOTICE: Each image can only display text for the mean, standard deviation, and box area for three images at a time. To view the data for a particular box, select the box on the image and the system displays the data for the box number selected.

**Figure 2 20cm QA Phantom Brightness Uniformity & CT# Measurement – Building
Placing Reference ROI Boxes**

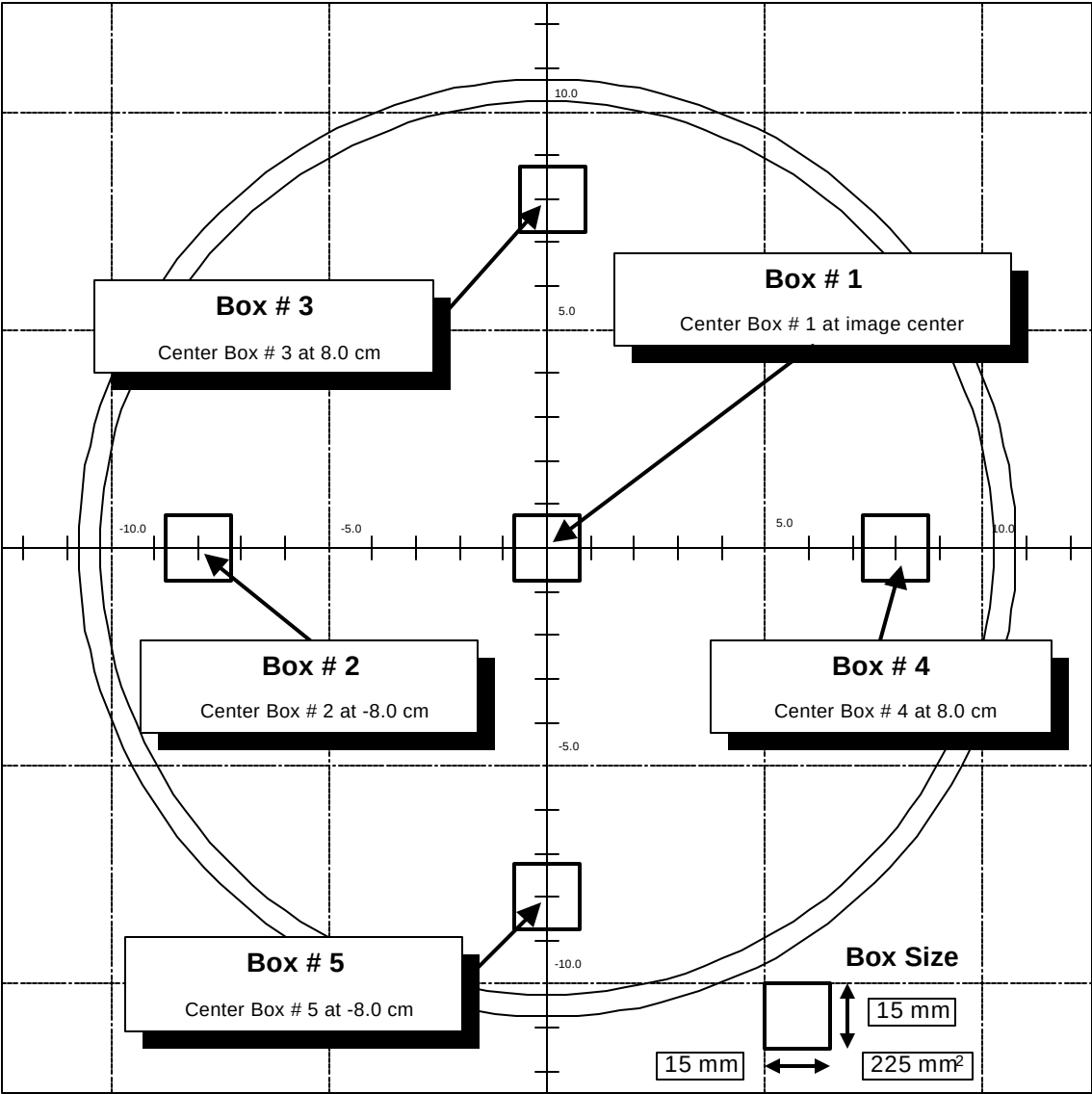


Table 5 **20cm QA Phantom CT# Brightness Uniformity & CT# Image Performance Worksheet**

Image	Box 1 (Center)	Box 2 (Left Center)		Box 3 (Top Center)		Box 4 (Right Center)		Box 5 (Bottom Center)		Outside Boxes Avg	
	Means (Xc)	Means	Std dev	Means	Std dev	Means	Std dev	Means	Std dev	Means (AvXo)	Std dev (AvSDo)
1			N.A.		N.A.		N.A.		N.A.		N.A.
2			N.A.		N.A.		N.A.		N.A.		N.A.
3			N.A.		N.A.		N.A.		N.A.		N.A.
4			N.A.		N.A.		N.A.		N.A.		N.A.
5			N.A.		N.A.		N.A.		N.A.		N.A.
6			N.A.		N.A.		N.A.		N.A.		N.A.
7			N.A.		N.A.		N.A.		N.A.		N.A.
8			N.A.		N.A.		N.A.		N.A.		N.A.

Box Size = 196 mm² to 256 mm²
 15 mm (+/- 1 mm) x 15 mm (+/- 1 mm)
 31 (+/- 2 pixels) x 31 (+/- 2 pixels)

Box Positions: Box 1 = 0 mm x 0 mm
 Box 2 = 0 mm x -80 mm
 Box 3 = 80 mm x 0 mm
 Box 4 = 0 mm x 80 mm
 Box 5 = -80 mm x 0 mm

Table 6 **20cm QA Phantom CT# Brightness Uniformity & CT# Row Performance Worksheet**

Row	Image s	Center Box Means (Xc)	Outer Boxes Means Averages (AvXo)	Outer Boxes Means Averages minus Center Box Means (AvXo – Xc)	Outer Boxes Standard Deviation Averages (AvSDo)	Comments
4A	1				N.A.	
3A	2				N.A.	
2A	3				N.A.	
1A	4				N.A.	
1B	5				N.A.	
2B	6				N.A.	
3B	7				N.A.	
4B	8				N.A.	
Specifications		0.00 +/- 3.0	N.A.	< +/- 3.0	N.A.	

4. Calculate the Brightness Uniformity and CT# values for each image in the series, record values in Table 6, and compare to specifications.
 - a. Calculate and record the average means values (X_o) for the four outside Boxes (Boxes 2 through 5) for each of the images and record in Table 5.
 - b. Record the center box (Box 1) means values (X_c) for each image in Table 3.
 - c. Record the outside boxes (Box 2 through 5) means average values (AvX_o) for each image in Table 6.
 - d. Calculate the Brightness Uniformity ($AvX_o - X_c$) value for each image and record in Table 6.
 - e. Verify Brightness Uniformity ($AvX_o - X_c$) value and CT# value (X_c) for each image meets specifications listed in Table 6.
 - f. Record the 20cm QA Phantom Brightness Uniformity ($AvX_o - X_c$) value and CT# value (X_c) for each image in the HHS Record Tables.

20CM QA PHANTOM IMAGE SERIES IMAGE PERFORMANCE FAILURE RECOMMENDED ACTIONS

SPECIFICATIONS:

Each Row (4A, 3A, 2A, 1A, 1B, 2B, 3B, and 4B) of the series must pass 20cm QA Phantom Brightness Uniformity and CT# (for the first series scan parameters) specifications:

- **$AvX_o - AvX_c$:** < +/- 3.0
- **X_c :** < +/- 3.0

RECOMMENDED RECOVERY:

If the specifications are not met, perform DAS Gain Calibration, Collimator Calibration, Detailed Calibration and Auto CT Number Adjust as detailed in Section 3.2.7 of this document. Repeat this procedure to verify 20cm QA Phantom Image Performance.

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